3.1, 10, 14, 24, 25

Problem 3.1. Use Table 3.1 to compute the temperatures of solid A and solid B when $q_A = 1$. Then compute both temperatures when $q_A = 60$. Express your answers in terms of ϵ/k, and then in kelvins assuming that $\epsilon = 0.1$ eV.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_{N,V} \qquad q_A = 1 \qquad \frac{S_A}{k} = 5.7 \qquad \frac{S_B}{k} = 186.4$$

$$U = q \cdot \epsilon \Rightarrow U(q=1) = 1 \cdot \epsilon$$

$$T_A = \frac{\partial U_A}{\partial S_A} = \frac{\Delta U_A}{\Delta S_A} = \frac{1\epsilon - 0\epsilon}{5.7 k_B - 0} = \frac{\epsilon}{5.7}$$

$$T_B = \frac{\Delta U_B}{\Delta S_B} = \frac{99\epsilon - 100\epsilon}{(186.4 - 187.5)k_B} = \frac{\epsilon}{1.1 k_B}$$

$$T_A < T_B$$

$$T_A = \frac{\epsilon}{5.7 \cdot k_B}$$

$$\epsilon = 0.1\,eV$$

$$T_A = \frac{0.1\,eV}{5.7 \cdot 8.6 \cdot 10^{-5}\,eV/K}$$

$$\rightarrow \boxed{T_A = 204\,K}$$

$$T_B = \frac{0.1}{1.1 \cdot 8.6 \cdot 10^{-5}}$$

$$\rightarrow \boxed{T_B = 1057\,K}$$

$$k_B = 1.38 \cdot 10^{-23}\,J/K$$

$$1eV = 1.6 \cdot 10^{-19}\,J$$

$$k_B = 8.6 \cdot 10^{-5}\,eV/K$$

$$T_A = \frac{\epsilon}{(159.1 - 157.4)k_B} = \boxed{\frac{\epsilon}{1.7\,k_B}}$$

$$T_B = \frac{-\epsilon}{(105.3 - 107.0)k_B} = \boxed{\frac{\epsilon}{1.7 k_B}}$$

$T_A = T_B$ ← at equilibrium

$$\boxed{T_A = 684\,K}$$

Problem 3.10. An ice cube (mass 30 g) at 0°C is left sitting on the kitchen table, where it gradually melts. The temperature in the kitchen is 25°C.

(a) Calculate the change in the entropy of the ice cube as it melts into water at 0°C. (Don't worry about the fact that the volume changes somewhat.)

(b) Calculate the change in the entropy of the water (from the melted ice) as its temperature rises from 0°C to 25°C.

(c) Calculate the change in the entropy of the kitchen as it gives up heat to the melting ice/water.

(d) Calculate the net change in the entropy of the universe during this process. Is the net change positive, negative, or zero? Is this what you would expect?

← latent heat ice → water $Q = 333.7\ J/g \cdot 30\ g = L_H \cdot m$

a. $\boxed{\Delta S = \frac{Q}{T}} = \frac{333.7 \cdot 30}{273.15} = 36.6\ J/K$